Characterization of Electromagnetic Fields in a GTEM Cell Using a KSQ1000 Comb Generator and Analysis of Comb Generator Technologies

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***Abstract*—This paper presents a comprehensive investigation into the characterization of electromagnetic fields within a Gigahertz Transverse Electromagnetic cell using a commercially available KSQ1000 comb generator. The measurements provide detailed information on the field uniformity and frequency response across the cell volume. This data is crucial for ensuring the accuracy and reliability of various electromagnetic compatibility testing applications conducted within the GTEM cell. Furthermore, the paper delves into the underlying technologies employed for comb generator manufacturing, with a specific focus on Non-Linear Transmission Lines. This analysis offers valuable insights into the selection criteria for comb generators.**

***Keywords*—electromagnetic fields, GTEM cell, comb generator, non-linear transmission line, electromagnetic compatibility**

I. INTRODUCTION

In the realm of electromagnetic compatibility (EMC) testing, the Gigahertz Transverse Electromagnetic (GTEM) cell serves as a critical tool for assessing the electromagnetic performance of electronic devices and systems. As the demand for reliable and accurate EMC testing continues to be important, there is the need for standardized measurement environments. This challenge is accentuated by the variety of GTEM cells available, each exhibiting unique design features and performance characteristics.

This paper delves into the intricate landscape of GTEM cells, focusing on the analysis of measurements obtained from two distinct cells. The main point of this investigation is the utilization of a comb generator as the primary source for measurements, a technique recognized for its precision and repeatability. By employing this measurement approach, the aim is to unravel the nuances of GTEM cell performance, shedding light on the key parameters that influence measurement outcomes.

Expected findings aim to contribute to the ongoing discourse surrounding standardized EMC testing methodologies, fostering a more robust and reliable framework for evaluating the electromagnetic compatibility of modern electronic systems.

This study incorporates the IEC Standard [1] as a guiding framework for evaluating the GTEM cells performance. It is recognized for its comprehensive guidelines on EMC testing methodologies, provides a structured approach to assessing the functionality and accuracy of EMC test environments.

Furthermore, the comb generator technology and its possible variations are discussed. To properly dissect the inner workings of such a device circuit simulation was conducted and analyzed. The simulations were done as a basis for a future prototype of a comb generator that could be used for more measurements regarding the analysis of GTEM cells and other measuring environments. Combining the planned manufacturing of the generator and conducting more complex measurements should result in a greater understanding of EMC testing in GTEM cells.

II. MEASUREMENT SETUP OVERVIEW

This study analyzed the measurement results from two different GTEM cells for characterizing radiated emissions from the comb generator, following the guidelines set forth in the standard [1].

Key equipment:

- radiated emissions source: KSQ1000 comb generator,
- EMI receiver: Gauss Instruments Tdemi m+,
- GTEM Cell 1: GTEM 500 manufactured by Astat,
- GTEM Cell 2: GTEM 5317 manufactured by Lindgren.

The KSQ 1000 comb generator served as the source of the reference signal for the radiated emission measurements, compliant with the guidelines outlined in [1]. With accordance to the mentioned guidelines the comb generator was set up in the GTEM cells as an EUT. While in the chamber the generator was the source of radiated emissions which were then measured with the EMI receiver.

The entire measurement process was repeated for the other GTEM cell, ensuring the EUT was subjected to identical test conditions.

The EMI receiver recorded the field strength level throughout the frequency range. This data was then used to analyze the radiated emission profiles obtained from each GTEM cell. The KSQ1000 comb generator provides a well-defined, broadband signal, ideal for characterizing the frequency response of the GTEM cells, as recommended for calibration purposes.

The Gauss Tdemi m+ EMI receiver offers high sensitivity and resolution, crucial for capturing subtle differences in the radiated emission profiles.

III. ANALYSIS OF THE MEASUREMENTS RESULTS

While GTEM cells offer a controlled environment for radiated emission testing, potential variations between cells could lead to inconsistencies in results. This can create uncertainty regarding a device's true compliance with regulatory standards. This comparison aimed to assess the

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level of agreement between the two cells in replicating a known radiated emission profile.

By analyzing the response of the cells to a well-defined signal source it is possible to gain insights into these potential factors and their impact on measurement accuracy.

All measurement techniques have inherent uncertainties. Analyzing the results from different cells helps to quantify the level of uncertainty associated with radiated emission measurements in GTEM cells. This knowledge allows for a more informed interpretation of test results.

The chosen signal source for this study was a comb generator with discrete frequency steps every 10 MHz, spanning a range of 30 MHz to 1 GHz. This selection provided a well-defined, broadband signal ideal for characterizing the frequency response of the GTEM cells. The analysis of the measured data revealed distinct characteristics in the two cells.

The results obtained for GTEM 500 cell are characterized by higher field strength level, particularly at lower frequencies which can be seen in Fig. 1. Additionally, the response displayed more variation across the measured frequency range. This might be due to factors like cell resonances or reflections at lower frequencies.

The results obtained for the GTEM 5317 cell showcase a lower field strength level with a fewer differentiations between the maximum values throughout the entire frequency range, which can be seen in Fig. 2.

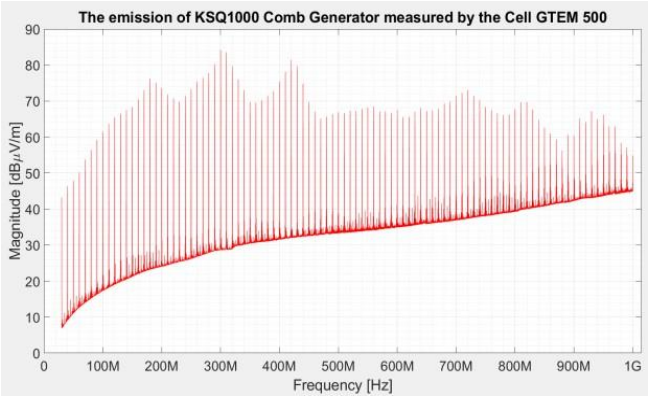


Fig. 1. The emission of KSQ1000 Comb Generator measured by the Cell GTEM 500

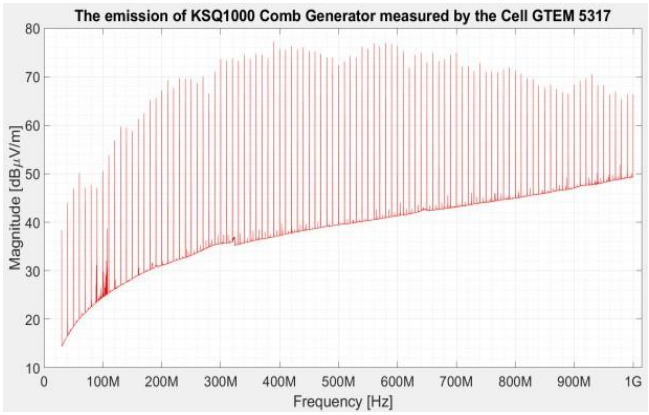


Fig. 2. The emission of KSQ1000 Comb Generator measured by the Cell GTEM 5317

The observed differences highlight the importance of considering cell specific characteristics when interpreting radiated emission measurements for regulatory compliance or product development. These findings warrant further investigation, such as:

- repeating the measurements with the same comb generator in both cells, which can confirm the observed trends and identify any random variations in cell response,
- analyzing the cell response to different signal sources beyond the comb generator can provide a broader understanding of their behavior across various radiated emissions. This could involve using broadband noise sources or signals with specific modulation schemes,
- performing a correlation analysis between the results from both cells can quantify the level of agreement and identify any systematic biases,

By conducting these additional studies, it is possible to gain a more comprehensive understanding of the factors influencing radiated emission measurements in different GTEM cells.

This analysis does not aim to compare the GTEM cells, but to accentuate the importance of further research with the use of reference source. That should include not only repeating the tests, but also conducting new ones with the use of open-area test sites and semi-anechoic chambers.

IV. Comb Generator Technologies Overview and Simulations

A. *Technologies overview*

Comb generators are essential tools for calibration and EMI testing. They produce a series of harmonically related frequencies, spaced at a constant interval, resembling a comb. The frequency range and harmonic level of a comb generator depends on its technology and input parameters.

The most common technology used was an pulse generator that would change the source signal and magnify it with amplifiers. For the source signal was used either an oscillator or a square wave generator.

Another idea for the comb generator construction were step recovery diodes (SRDs). When subjected to a step voltage, SRDs generate sharp, narrow pulses that can be Fourier transformed into a comb of harmonics

Lately some developers decided to use for the comb generator non-linear transmission lines (NLTL). The idea is simple, it uses high impedance transmission lines and non-linear elements to produce short pulses. Normally NLTLs use elements like capacitors, but for the specific use of comb generators most common are diodes, since they can emit far faster responses. This technology will likely evolve, since it offers potential for optimization. Already there can be seen ideas of using semiconductors and monolithic microwave integrated circuits to broaden the frequency range.

While the basic NLTL structure is relatively simple, circuit simulation can be complex due to optimization requirements. As NLTL technology shows promise for comb generator development, this paper discusses its circuit simulations.

B. *Simulations*

The NLTL technology was chosen for circuit simulation of a comb generator circuit. The idea was to create a basic circuit and observe the output signal and harmonics it would produce. Since the main part of the circuit are inductances and diodes the optimization of the circuit was based on checking different parameters. The circuit that was simulated can be observed in Fig. 3.

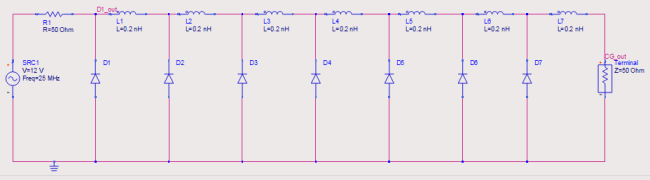


Fig. 3. The circuit simulation of NLTL based comb generator

It uses elements that are commercially available like an oscillator, Schottky diodes and coils. The signal output is observed on the node of the terminal. The terminal acts as a represantation of a connector. In an actual device it would be an antenna or a wire connection. The output signal and the harmonics can be observed in Fig. 4 and Fig. 5 respectively.

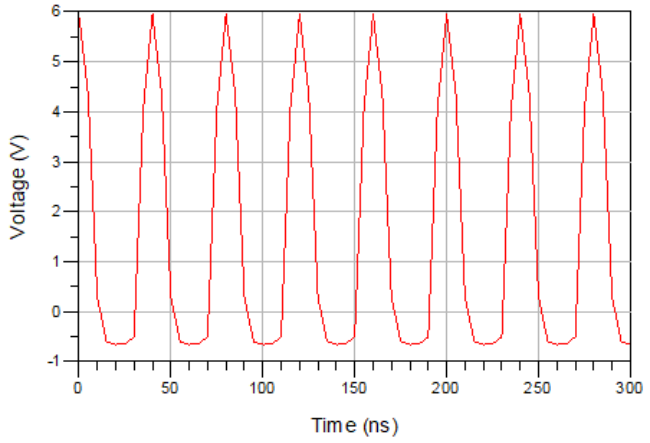


Fig. 4. The signal produced by the simulated circuit

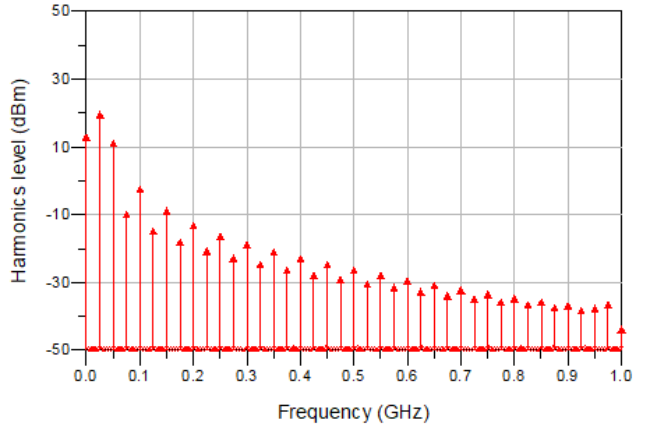


Fig. 5. The harmonics produced by the simulated circuit

The simulation that was performed in the Advanced Design Simulation (ADS) software used a harmonic balance simulation engine. Harmonic balance is a powerful tool for analyzing how nonlinear circuits and systems distort signals, particularly in the realm of RF and microwave engineering where frequency-domain analysis is key. It can simulate a wide variety of components like amplifiers, mixers, and modulators under large input signals.

One of its strengths is handling multi-tone simulations, meaning it can capture the complex interactions between multiple frequencies in a circuit, including the generation of harmonics and intermodulation products, not just from the circuit itself, but also from the input signals.

The technique works iteratively, assuming that any steady-state response to a sinusoidal input can be represented by a finite Fourier series. It involves a back-and-forth between the time and frequency domains: currents through linear elements are calculated in the frequency domain, while those through nonlinear elements are calculated in the time domain and then transformed to the frequency domain.

The truncated Fourier approximation, combined with the circuit equations, results in a residual function that is minimized. This leads to a system of nonlinear algebraic equations that are solved for the Fourier coefficients using Newton's method. The inner linear problem within this process is solved using either a direct method (Gaussian elimination) for small problems or a Krylov-subspace method (e.g., GMRES) for larger problems.

Finally, nonlinear devices like transistors are evaluated in the time domain and their behavior is then converted to the frequency domain using the Fast Fourier Transform (FFT).

Using the discussed tool allowed to produce a signal and harmonics based on the designed circuit. This tool used a frequency sweep to calculate results with 200 samples over the range of 1GHz. That meant that every 5MHz the circuit was tested. With that it was possible to view the harmonics from the simulated circuit. For the voltage signal additional tool was used that through a reverse FFT was able to show the voltage signal on the load of the circuit.

The produced signal is a series of pulses with a rise time of 10 ns. The voltage level was around 6 V which was half of the voltage input. The pulse was generated every 40 ns. Depending on the input signal frequency and amplitude the rise time can change.

The harmonics produced are showing the correct step in between. The comb generator should produce harmonics over a desired frequency range with a certain step that is decided by the input frequency. With the use of 25 MHz oscillator the step of the same value is acquired. The main issue with the presented results is that only 3 harmonics are over the 0dB power level. That is because of a signal rise time being around 10 ns. With faster rise time the harmonic range should be greater The view of the harmonics was generated by the ADS software and harmonic balance engine and no window functions were added onto it.

C. *Further analysis*

The obtained circuit needs to be optimized more. The ideal situation would be to create a circuit where not only the diodes and inductance parameters are taken into consideration, but also the spacing of the sections. Also what needs to be found is a diode that would result in much faster responses. 10ns of rise time is not a fast enough value, since it is a parameter responsible for a frequency range. In this case 1GHz range is what was the aim, thus achieving rise times lower than 1ns will be the next research goal.

Creating a more complex simulation of the circuit could bring more information about the NLTL technology used for comb generators. Instead of an output terminal it would be good to simulate an antenna output or to design a close representation of it.

The harmonics level could be higher, for which the circuit would have to be more complex. For example, certain frequency filters could be added or mentioned before width and length of the sections. In theory adding more sections should affect the frequency range and thus the harmonics level, but it wasn't observed through the simulations. Which possibly is because of not enough factors were considered.

The achieved circuit and simulation results are promising and based on them it should be possible to manufacture a working prototype of a comb generator that can be used on further characterization of electromagnetic fields in GTEM Cells and other measurement environments.

V. SUMMARY

This paper discussed the measurements done in GTEM Cells with the use of KSQ 1000 comb generator as a basis for the analysis of differences between those cells. It's not possible to conclusively state the reasons between those discrepancies at this time, but there are solid basis for further investigation.

Furthermore, the simulations of the comb generator circuit still need to be optimized, but they give opportunity to manufacture actual device that could be used for further research with GTEM cells and other environments like anechoic chambers.

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